

QF620 Stochastic Modelling in Finance

Assignment 4/4

Due Date: 6-Nov-2024

1. A contract pays (on maturity date T)

$$V_T = \sqrt{S_T}.$$

Derive a valuation formula for the contract at time 0 using

- (a) Black-Scholes model

$$dS_t = rS_t dt + \sigma S_t dW_t^*.$$

- (b) Static replication approach.

$$\int_0^F h(K) \frac{\partial^2 P(K)}{\partial K^2} dK + \int_F^\infty h(K) \frac{\partial^2 C(K)}{\partial K^2} dK$$

② Black-Scholes model:

$$dS_t = rS_t dt + \sigma S_t dW_t^*$$

$$\text{let } F = S_0 \cdot e^{rT}$$

$$V_0 = e^{-rT} \cdot E(\sqrt{S_T})$$

$$\therefore S_T = S_0 \cdot e^{((r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}Z)}$$

then, $Z \sim N(0, 1)$

$$\therefore \sqrt{S_T} = \sqrt{S_0} \cdot e^{(\frac{1}{2}(r - \frac{\sigma^2}{2})T + \frac{\sigma\sqrt{T}}{2}Z)}$$

$$E(e^{cZ}) = e^{\frac{c^2}{2}}$$

$$\text{let } \frac{\sigma\sqrt{T}}{2} = c,$$

$$\therefore E(e^{\frac{\sigma\sqrt{T}}{2}Z}) = e^{\frac{(\frac{\sigma\sqrt{T}}{2})^2}{2}} = e^{\frac{\sigma^2 T}{8}}$$

$$\text{plug } E(e^{\frac{\sigma\sqrt{T}}{2}Z}) = \frac{\sigma^2 T}{8} \text{ into } E(\sqrt{S_T})$$

$$\therefore E(\sqrt{S_T}) = \sqrt{S_0} \cdot e^{(\frac{1}{2}(r - \frac{\sigma^2}{2})T + \frac{\sigma^2 T}{8})}$$

$$\begin{aligned}
 \therefore V_0 &= e^{-rT} \cdot \sqrt{S_0} \cdot e^{\left(\frac{1}{2}(r-\frac{\sigma^2}{2})T + \frac{\sigma^2 T}{8}\right)} \\
 &= e^{-rT} \cdot \sqrt{S_0} \cdot e^{\frac{1}{2}rT - \frac{1}{4}\sigma^2 T + \frac{1}{8}\sigma^2 T} \\
 &= e^{-rT} \cdot \sqrt{S_0} \cdot e^{\frac{1}{2}rT - \frac{1}{8}\sigma^2 T} \\
 \therefore V_0 &= \sqrt{S_0} \cdot e^{\left(-\frac{rT}{2} - \frac{\sigma^2 T}{8}\right)}
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{b} \quad h(S_T) &= \sqrt{S_T} = S_T^{\frac{1}{2}}, \quad h'(S_T) = \frac{1}{2} S_T^{-\frac{1}{2}} = \frac{1}{2\sqrt{S_T}} \\
 h''(S_T) &= \frac{1}{2} \cdot \left(-\frac{1}{2}\right) S_T^{-\frac{3}{2}} = -\frac{1}{4} S_T^{-\frac{3}{2}} = -\frac{1}{4\sqrt{S_T}^3}
 \end{aligned}$$

$$\begin{aligned}
 V_0 &= e^{-rT} h(F) + \int_0^F h''(k) \cdot P(k) dk + \int_F^\infty h''(k) C(k) dk \\
 &= e^{-rT} \cdot \sqrt{S_T} + \int_0^F \frac{1}{4\sqrt{k}^3} \cdot P(k) dk + \int_F^\infty \frac{1}{4\sqrt{k}^3} C(k) dk \\
 &= e^{-rT} \cdot \sqrt{S_0} \cdot e^{\frac{rT}{2}} - \int_0^F \frac{1}{4\sqrt{k}^3} \cdot P(k) dk - \int_F^\infty \frac{1}{4\sqrt{k}^3} C(k) dk \\
 &= e^{-rT} \cdot \sqrt{S_0} \cdot e^{\frac{rT}{2}} - \int_0^F \frac{1}{4\sqrt{k}^3} \cdot P(k) dk - \int_F^\infty \frac{1}{4\sqrt{k}^3} C(k) dk \\
 &= \sqrt{S_0} \cdot e^{-\frac{rT}{2}} - \int_0^F \frac{1}{4\sqrt{k}^3} \cdot P(k) dk - \int_F^\infty \frac{1}{4\sqrt{k}^3} C(k) dk
 \end{aligned}$$